



Capstone Proposal

Capstone should challenge your abilities, pushing you to explore new research and development territories. The project should align with your prospective career path and personal passions, offering a valuable addition to your portfolio. Its core objective is to exhibit your current proficiency while showcasing your adeptness in mastering new technologies through a creative and technically impressive project.

Name: Lily McClellan

Capstone Title: Causerie

Project Summary

Please provide a summary of your proposed Capstone project from a development perspective. Explain what you are building, the primary systems and features involved, the technologies and tools you plan to use, and the major technical challenges you expect to address. Identify the primary technical challenge, research area, or area of innovation that makes this project worthy of Capstone status and explain how it extends beyond work completed in previous courses. Focus on the implementation of the project rather than its story, theme, or content.

Your response should be between 150 and 250 words.

The term 'causerie' refers to an intellectual informal discussion. Often employed in language learning settings, it's often used for encouraging low stress practice of a target language. That is exactly what this project will provide. Causerie is an AI driven dialogue application packed with research backed features aimed at allowing safe and controlled studying. Key features of this application include topic-based dialogues, a sticky note notation system, goals/progress tracking, targeted corrections/practice, and a reminder system for building study habits. Visually, this app will have a soft and welcoming UI and a simple mascot to give a personable aspect to dialogues.

Causerie will be a local desktop application aimed at portability and cross-platform compatibility, allowing support for both local Ollama model integration or external AI API integration. The tech stack is as follows:

- Tauri2 (Software Framework)
- React/TS/Vite (Front-End)
- PixiJS (Character Rendering)
- XState (State Machine)
- SQLite (Database)
- Go (Back-End)

The strongest challenge anticipated with Causerie is state tracking. Three very different technologies need to work together in a delicate system (React's declarative design, PixiJS's imperative design, and Go's business functionality). XState solves this issue of state management. In addition to the technical challenge, I am ensuring that all features align with real linguistic research, engaging direct/indirect learning strategies and intentional design to be as welcoming and educational as possible. This extends beyond prior coursework in four technical pillars: adaptive/predictable state flows, cross-platform compatibility, research-driven targeted design, and agentic design flow.

Motivation

Explain the inspiration and rationale behind your project selection, including your personal, professional, or technical interests in the project. Discuss what you hope to achieve through this experience and why the project represents an appropriate Capstone challenge. Consider how the project aligns with your career goals, expands your skills, or explores opportunities within existing technologies, products, or industries.

Your response should be between 100 and 150 words.

If you too have ever studied a foreign language or have a second language that isn't spoken by most of your family, you are familiar with the biggest bottleneck that many learners experience: lacking an environment to practice and learn in. One of the most effective ways to learn a language is to *use it*, but so many people cannot. I hope to make a project that gives intermediate and advanced learners who are caught in a state of 'learning limbo' a place to sharpen and practice skills. Technically, Causerie will provide me with vital full-stack experience and an opportunity to create a product that fulfills a genuine market need. With the complex interlocking pieces, I will have the opportunity to work on a much different scale than I have prior done. Ultimately, I feel Causerie is a perfect professional demonstration of and challenge to my repertoire of skills.

Technologies and Research Requirements

Identify the technologies, tools, frameworks, hardware, software, or platforms you plan to use in your Capstone project. For each item, provide a brief explanation of its purpose and how it will contribute to the project.

Present your response as annotated bullet points. Clearly indicate which technologies are new to you and will require additional research, experimentation, or skill development.

Example:

- **Unreal Engine 5** – Primary game engine used to develop gameplay systems, rendering, and user interfaces. I have prior experience with Unreal and do not anticipate significant research requirements.
- **OpenXR** – Used to provide VR headset compatibility. This technology is new to me and will require research into VR interaction systems and device integration. I plan to learn through official documentation, tutorials, and prototype development.

Technologies

- **Tauri2** – Tauri2 will be used as the primary software framework that the application is built on. It allows for the use of standard web technologies while allowing privileged operations behind the Go layer.
 - I do not have experience with Tauri, but I have some experience with a similar technology: Electron. Tauri was chosen over Electron for its lower memory footprint, which is beneficial for applications that are meant to be open for extended periods of time. I plan to learn Tauri through the official documentation as well as with AI assistance.
- **React** – React is the main framework for developing the front-end of the application, which was chosen for its versatility in developing dynamic interfaces as well as its place as a standard choice in industry level projects.
 - I am familiar with React, having used it in other projects, but will be taking my skills further than I had before in tying React modules/components to the state machine.
- **TypeScript** – TypeScript serves as the front-end scripting language for developing React components. It is a natural choice for the project, having all the benefits of being an integrated web technology while retaining type safety. TypeScript will also be responsible for communication between React/Vite and PixiJS/XState.
 - I have a lot of experience in using TypeScript, but will need to do additional research for developing React components.
- **Vite** – Vite is the primary framework for styling the application, having a rich ecosystem when paired with the aforementioned technologies.
 - I have limited experience with Vite, and will need to do more research to develop my own modules/components.

- **PixiJS** – PixiJS is an HTML5 creation engine specialised in 2D rendering through WebGL. This component is for rendering the mascot of the application, which is necessary due to its complex visual and programmatic states.
 - I have never used PixiJS before nor WebGL, and will be learning this technology from scratch. I plan to learn it via official documentation, example projects, and AI assistance.
- **XState** – XState is a state management solution that naturally integrates with TypeScript. XState provides an event-driven approach to handling state changes across the front-end components.
 - I have never used XState, and have never used complex state management solutions before, but have a strong understanding of its use cases and development cycle. I will develop my knowledge of this technology via official documentation, example projects, and AI assistance.
- **SQLite** – SQLite is a lightweight database built into most platforms/frameworks. It will be used for local data storage for handling persistence of states, user preferences, progress tracking, etc.
 - I have extensive experience with SQLite and adjacent technologies, and I am very confident in my ability to develop with it.
- **Go** – Go will serve as the back-end scripting language for creating the main business logic. This includes API calls, system level state management, and behind-the-scenes application logic.
 - I have a good amount of experience in programming in Go as well as developing business side logic in larger applications with it. I will only need to do research into integrating it with React.
- **OpenAI API** – The primary AI API that the application will be developed to use. OpenAI provides an affordable plan that gives ample headroom for development.
- **DeepSeek API** – The secondary AI API that the application will use is DeepSeek, which is cheaper and comes with less advanced reasoning, making it a good back-up API and additional API to test that API calls function similarly across platforms.

Tools

- **WebStorm** – WebStorm is a JetBrains IDE specialised in web-based development, making it an excellent choice for developing the front-end of the application. WebStorm does not directly support PixiJS or XState, but there are community plugins available to assist with this.
- **GoLand** – GoLand is a JetBrains IDE specialised in developing Go projects. I will use it for developing the back-end of the application, and will be using built in testing features to ensure the back-end and API side functions as expected
- **Codex** – Codex will be my agent of choice for enabling agentic development of this project, allowing me to develop more efficiently on this project.
- **Aseprite** – Aseprite is a digital art software specifically targeted at Pixel art, which will be the primary style of the mascot for the project. This software may not be used if I choose to source my art externally.

Platforms

- **Linux** – Causerie will primarily be developed on Linux, consequentially making the best product the Linux bundled version.

- **Windows/MacOS** – Causerie will not be developed on these platforms, but they will be involved in the testing process to ensure expected cross-platform functionality.

Minimum Viable Product (MVP)

The **Minimum Viable Product (MVP)** is the smallest version of your project that still demonstrates its primary goals and core technical challenges. If you were forced to stop development at the minimum acceptable point, what functionality would need to be completed for the project to still be considered a Capstone-level achievement?

List the required features, systems, and functionality that must be completed. Focus on what the project must do, not on optional enhancements or future ideas.

Examples:

- The game contains a complete gameplay loop with win and lose conditions.
- The game can save and load player progress.
- The game creates a unique dungeon layout each time a new game is started.

Initial Set-Up

- On initial launch, the application should run an initial set-up. Initial set-up should have the user select a target language, proficiency level (including 'unknown'), and frequency of reminders (options for what time of day and if daily/what days of the week).
- After initial launch, the application should run an introductory conversation about introducing yourself. This asks for the user's name, gender, and preferred areas of practice. This will be programmed, and will not involve the AI chatbot.
- The application should (after set-up) allow the user to create threads and give them a name and topic.

Threads

- Threads should each have their own stored contexts to guide the chatbot in responses (alongside conversation history).
- Threads can either be initialised with a user prompt or a generated prompt from the AI.
- Creating a thread will open up a thread workspace. This includes a section for the mascot, a section for the conversation history, and a section for the user to enter responses.

Character/Mascot

- The mascot will minimally change states between greeting (initial prompt), idling (time without prompt), thinking (awaiting API response), and talking (outputting API response)
- The mascot should provide user-friendly error feedback for common errors (API cannot be reached, API limit reached, internal error, etc)
- Mascot should have an artistic representation for the aforementioned states and be able to transition between them as states update.

Sticky Notes

- During conversations, the user can highlight (a) word/s and select a sticky option, which will add a sticky note to the highlighted section.
- Sticky notes colours can be selected on creation/edit. Stickies appear as a clickable highlight overtop words wherever they appear in chatbot responses.

- Sticky notes can store a title (shown when hovering over a stickied phrase) and a body (shown when clicking on a highlight).
- Stickies can be deleted via button click.
- Sticky will have at least one placeholder value (chosen at random if multiple) showing the user what an effective note looks like (engaging cognitive strategy– NOT writing translations)

Hints

- During conversation, a user can highlight (a) word/s and select a hint option, which opens up a menu that shows the dictionary definitions of each of the highlighted words.
- During conversation, a user can highlight (a) word/s and select a translate option, which prompts the mascot to provide a brief translation of the section (not recorded to conversation history)

Reminders

- The application should send push notification reminders at the defined dates/times.
- Push notification reminder should include an encouraging remark and on click should open the application.
- The application should register notification schedules as OS-level notifications to ensure they are delivered regardless of the app state.

Goals

- User should be able to define up to three learning goals (such as “I want to expand my vocabulary” or “I want to work on my grammar”) which the AI will focus on when providing feedback
- Goals should minimally be selectable from a list.

Technical

- The application can save and update user preferences.
- The application can be run on Linux, Windows, and MacOS with minimal disruptions.
- The application will ask for an OpenAI API key (MVP only supports this AI method).
- The application will manage LLM context carefully to ensure modular verifications and checks without context bleeding.

Milestones and Timeline

Provide a tentative timeline for your Capstone project. While project plans may evolve throughout development, this timeline should represent your current expectations and demonstrate that you have considered the major phases of production.

Include key milestones such as:

- Research, Planning and Documentation
- Prototyping
- Core Development
- MVP Complete
- Testing and Refinement
- Final Presentation and Submission

Present your timeline in a table and include the major phases of development along with the estimated amount of time required to complete each phase. While timelines may change as the project evolves, your timeline should reflect your current expectations for completing the Capstone.

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Faculty Review

Proposal Approved: ☐ Yes ☐ No

Faculty Comments: